

Complements of Operational Research

4.2 Replacement theory: with the time value of money

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1 € today is not worth 1 € five years from now

In 4.1 we added up costs from different years as if they were worth the same. But the company could invest the money (or pays interest on what it borrows) at a rate r per year.

Discount factor

$$d = \frac{1}{1+r}$$

A cost C paid k years from now is worth $C d^k$ today (**present value**).

With $r = 10\%$: $d = 0.9091$; 1000 € five years from now are worth $1000 \times 0.9091^5 = 620.92$ € today.

Convention (the one used in class)

- the purchase P is paid at the start (time 0);
- the cost of year i , C_i , is paid at the **start** of that year $\rightarrow$ factor d^{i-1} ;
- the resale S_n is received at the **end** of year $n \rightarrow$ factor d^n .

Question: with $r = 10\%$, should the diesel van from 4.1 still be replaced after 5 years?

The criterion: discounted equivalent annual cost

Criterion

$$W(n) = \frac{P + \sum_{i=1}^n C_i d^{i-1} - S_n d^n}{\sum_{i=1}^n d^{i-1}}$$

Replace after n^* years, with $W(n^*)$ minimum.

With $r = 0$ ($d = 1$), $W(n)$ is the average cost $A(n)$ of 4.1.

Interpretation: $W(n)$ is the **constant annual payment** (at the start of each year) with the same present value as a cycle of n years.

Minimising $W(n)$ is the same as minimising the present value of *all* future cycles (van after van), because

$$\sum_{k \geq 0} \text{cycle} \cdot d^{kn} = \frac{\text{cycle}}{1 - d^n}$$

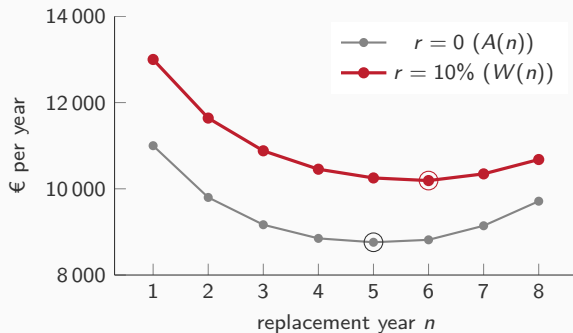
and $1 - d^n = (1 - d) \sum_{i=1}^n d^{i-1}$.

The van with $r = 10\%$

n	C_n	d^{n-1}	$C_n d^{n-1}$	$\sum C_i d^{i-1}$	$S_n d^n$	$P + \sum -S_n d^n$	$\sum d^{i-1}$	$W(n)$
1	3000	1.0000	3000.0	3000.0	20000.0	13000.0	1.0000	13000.0
2	3600	0.9091	3272.7	6272.7	14049.6	22223.1	1.9091	11640.7
3	4400	0.8264	3636.4	9909.1	10142.7	29766.3	2.7355	10881.4
4	5400	0.7513	4057.1	13966.2	7513.1	36453.0	3.4869	10454.4
5	6400	0.6830	4371.3	18337.5	5588.3	42749.2	4.1699	10251.9
6	7600	0.6209	4719.0	23056.5	4233.6	48822.9	4.7908	10191.0
7	9800	0.5645	5531.8	28588.3	3181.6	55406.7	5.3553	10346.2
8	12500	0.5132	6414.5	35002.8	2332.5	62670.3	5.8684	10679.2

$P = 30\,000$; S_n as in 4.1 (22000, 17000, ..., 5000). With discounting, the optimum moves to **6 years** ($W = 10\,191$ €/year). Without discounting it was 5 years.

Why discounting postpones replacement



r	n^*	$W(n^*)$
0%	5	8760.0
5%	6	9514.6
10%	6	10191.0
15%	6	10846.0

With discounting, the purchase (today, undiscounted) weighs **more** than the future (discounted) costs. It pays to space out the purchases: the optimum moves to the right.

The level of W rises with r : tied-up capital has a cost.

The rule with discounting

Rule

Keeping it one more year costs (in value at the start of that year)

$$M_{n+1} = C_{n+1} + S_n - d S_{n+1}.$$

Replace at the end of year n when $M_{n+1} > W(n)$.

With no resale (zero scrap value): $M_{n+1} = C_{n+1}$, and the rule becomes

$$C_n \leq W(n-1) \quad \text{and} \quad C_{n+1} > W(n).$$

With $r = 0$ it is the rule of 4.1.

For the van ($r = 10\%$)

- 6th year: $M_6 = 7600 + 9000 - 0.9091 \times 7500 = 9781.8 < W(5) = 10\,251.9 \rightarrow \text{keep}.$
- 7th year:
 $M_7 = 9800 + 7500 - 0.9091 \times 6200 = 11\,663.6 > W(6) = 10\,191.0 \rightarrow \text{replace at the end of the 6th year}.$

$S_n - d S_{n+1}$: what is lost by selling one year later (the resale value drops and arrives one year later).

```
def custo_equivalente(P, C, S, r):  
    d = 1 / (1 + r)  
    W, soma_c, soma_d = [], 0, 0  
    for n, (c, s) in enumerate(zip(C, S), start=1):  
        soma_c += c * d**(n - 1)    # paid at the start of year n  
        soma_d += d**(n - 1)  
        W.append((P + soma_c - s * d**n) / soma_d)  
    return W
```

```
P = 30000  
C = [3000, 3600, 4400, 5400, 6400, 7600, 9800, 12500]  
S = [22000, 17000, 13500, 11000, 9000, 7500, 6200, 5000]  
for r in (0, 0.05, 0.10, 0.15):  
    W = custo_equivalente(P, C, S, r)  
    n = W.index(min(W)) + 1  
    print(r, n, round(min(W), 1))  
# 0 5 8760.0 | 0.05 6 9514.6 | 0.1 6 10191.0 | 0.15 6 10846.0
```

In the spreadsheet: one column with d^{n-1} , another with $C_n d^{n-1}$, cumulative sums and the division.

Try it: from what rate r would the optimum move to 7 years?

Example 4

A machine costs 500 m.u.; the discount rate is 5% and the scrap value is 0.

Year	1	2	3	4	5
Running cost	0	100	200	300	400

When should the machine be replaced by an identical one? Compare with the answer without discounting.

Exercise sheet 4 — Exercise 2 (homework)

Replacement with discounting (homework)

A machine costs 60 000 m.u.; the discount rate is 10% and the scrap value is 0. The running costs are:

Year	1	2	3	4	5	6	7	8	9	10
Cost	10000	10000	10000	10000	10000	13000	16000	19000	22000	25000

When should the machine be replaced by an identical one?

Submission: by email to the lecturer, by the deadline given in class. All exercises: [Exercise sheet 4](#).

KEY TAKEAWAYS — 4.2

1. $d = 1/(1 + r)$; cost of year i with factor d^{i-1} , resale at the end of year n with d^n .
2. Discounted equivalent annual cost: $W(n) = (P + \sum C_i d^{i-1} - S_n d^n) / \sum d^{i-1}$; replace after n^* with W minimum.
3. Rule: replace when $C_{n+1} + S_n - dS_{n+1} > W(n)$ (no resale: $C_{n+1} > W(n)$).
4. Discounting tends to **postpone** replacement (for the van: $5 \rightarrow 6$ years with $r = 10\%$).

Next: 4.3 Equipment that fails suddenly — group replacement

References Full list: *Course bibliography*.

- Kasana, H. S., Kumar, K. D., *Introductory Operations Research*, Springer, 2004 — ch. 11 (replacement when the value of money changes).
- Taha, H. A., *Operations Research: An Introduction*, 10th ed., Pearson, 2017.
- Jardine, A. K. S., Tsang, A. H. C., *Maintenance, Replacement, and Reliability*, 2nd ed., CRC Press, 2013 — equivalent annual cost.

Resources

- Exercise sheet 4. `code/python/04_replacement/04_2_discounting/valor_temporal.py`.